Question:

An automobile battery produces a potential difference (or "voltage") of 12 volts between its terminals. (It really consists of six 2 volt batteries following one after the other.) A headlightbulb is to be connected directly across the terminals of the battery and dissipate 40 watts of joule heat. What current will it draw and what must its resistance be?

Answer Choices:

A. 3.0 amps, 4 ohms

B. 2.8 amps, 3.2 ohms

C. 3.75 amps, 3.2 ohms

D. 4 amps, 3 ohms

E. 3.33 amps, 3.6 ohms

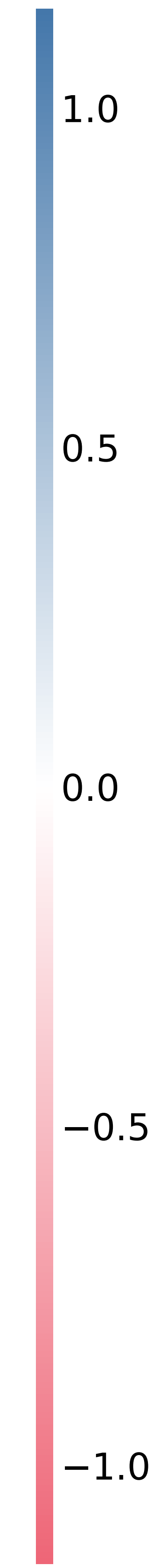
F. 5 amps, 2.4 ohms

G. 4.5 amps, 2.67 ohms

H. 2.2 amps, 5.45 ohms

I. 2.5 amps, 4.8 ohms

J. 3.5 amps, 4 ohms



Reasoning + Answer:

<think>

according to ohm's law, current is equal to voltage over resistance. Therefore, we can use the equation i = v/r, where i is the current, v is the voltage, and r is the resistance.

first, we need to calculate the resistance of the headlightbulb, we can use the equation r = v/i.

we are given that the power (p) is 40 watts and the voltage is 12 volts.

power is equal to current squared over resistance, so we can write p = i^2/r. Rearrange the equation to isolate r: r = i^2/p. therefore we can solve for r:

first of all we have to solve for i: i = p/v

hence simplify

from this i can be obtained </think><answer>3.5 amps</answer>

To obtain the answer, we must solve v

thus resistance = 3.5*3.5/12